ITMLA2-44 Project

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Table of contents

1. Project
2. Instructions to Students
3. Section A
    3.1. Question 1
    3.2. Question 2
    3.3. Question 3
    3.4. Question 4
4. Appendix

Project

Faculty:	Information Technology
Module Code:	ITMLA2-44
Module Name:	Machine Learning Algorithms
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Total Marks:	100
Submission Week:	Week 6

This module is presented on NQF level 6.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual project.

This project contributes 40% towards the final mark.

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
7. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

Section A

Learning Objective

To implement Machine Learning algorithms using Python in solving real-world problems.

Scope

Week 1 – 6

Technical Aspects

Submit both your jupyter notebook and documentation of your code where necessary.

Marking Criteria

Plagiarism is strictly forbidden. Any work found to be plagiarised will receive an immediate mark of 0.

AI Use Requirements for This Assessment

This assessment is classified as AI-1, as defined in the AI Use Category Table below.

AI Use Category Table

For this specific assessment, the following applies:

Code	Category	What This Means	Evidence Required	Disclosure Required
AI-1	Limited AI Use	You may use AI only for simple support tasks, for example, to explain Python errors, clarify machine learning concepts, assist with debugging, or improve the clarity of written explanations. However, the implementation, model outputs, evaluation, interpretation, and final discussion must reflect your own understanding and work. The main work should be your own.	In some tasks, you may also be asked for evidence of your process, for example, screenshots or extracts of AI prompts and responses used for debugging or clarification, a short explanation of how AI suggestions were checked or corrected and evidence that you understood and verified any AI-assisted changes before including them in your submission.	You must complete and submit an **AI Disclosure Appendix** explaining how you used AI. Access the AI Disclosure Appendix Template here.

Question 1 — 25 Marks

Study the scenario and complete the question(s) that follow:

Predictive Maintenance for Municipal Water-Pump Assets

A municipal water utility operates a large fleet of pumping assets across several service zones. Unexpected mechanical failures interrupt water supply, increase emergency repair costs and shorten the useful life of equipment.

You have been provided with historical telemetry, maintenance indicators and operating condition data for 2,149 pump assets. The target variable, Major_Service_Required, records whether an asset required a major service intervention during the observation period (1 = Yes, 0 = No).

Your task is to prepare the data and identify the most informative predictors that can later be used to build a predictive maintenance model. Use the dataset machine_service_risk.csv. Appendix A describes the dataset fields.

1.1 Prepare the dataset for machine-learning analysis. Your work must include an initial data audit, appropriate treatment of missing values, removal of non-predictive identifiers, encoding of categorical variables and feature standardisation. Briefly justify the key preprocessing decisions.

(8 Marks)

1.2 Apply SelectKBest with an appropriate classification scoring function to retain the eight strongest predictors of Major_Service_Required. Display the selected feature names and their scores.

(5 Marks)

1.3 Apply Recursive Feature Elimination (RFE) with Logistic Regression as the estimator to retain eight predictors. Display the selected feature names.

(6 Marks)

1.4 Compare the two selected feature sets. Identify the common predictors and explain what the agreement and differences between the two methods indicate about feature selection.

(6 Marks)

[Subtotal: 25 Marks]

End of Question 1

Question 2 — 30 Marks

Study the scenario and complete the question(s) that follow:

Maintenance Triage Model Selection

The water utility now wants a deployable model that can flag pump assets likely to need major service. Maintenance planners will use the prediction to prioritise inspections before a breakdown occurs.

Using the RFE-selected predictors from Question 1, compare a Logistic Regression classifier with a Decision Tree classifier. The utility is particularly concerned about false negatives because an asset that is incorrectly classified as low risk may fail before it is inspected.

2.1 Create a stratified 75/25 training and test split using the RFE-selected predictors. Train both a Logistic Regression classifier and a Decision Tree classifier. Use settings that make the models reproducible and suitable for the dataset.

(10 Marks)

2.2 Evaluate both classifiers on the same test set using accuracy, precision, recall, F1-score and a confusion matrix. Present the results clearly.

(12 Marks)

2.3 Recommend one of the two classifiers for maintenance triage. Support your recommendation using the recall results and the number of false negatives produced by each model and explain why this is important in the given maintenance context.

(4 Marks)

2.4 Before deploying the selected classifier, explain why its performance should not be judged using only this single train-test split. Describe one suitable method that could be used to obtain a more reliable estimate of how the model is likely to perform on unseen pump data.

(4 Marks)

[Subtotal: 30 Marks]

End of Question 2

Question 3 — 30 Marks

Study the scenario and complete the questions that follow:

Predicting Electricity-Restoration Time

A regional electricity distributor records operational information for past network incidents. Accurate restoration-time estimates help the control centre communicate with customers, dispatch the correct crews and manage escalation procedures.

The supplied dataset contains 1,000 historical incidents, including affected line length, fault category, grid-load level, dispatch window, crew type, dispatch delay and technician experience. The continuous target is Restoration_Time_Minutes.

Develop a Linear Regression model that learns its parameters using Gradient Descent implemented in Python. Do not use sklearn.linear_model.LinearRegression or SGDRegressor for the training step. Use grid_restoration.csv. Appendix B describes the fields.

3.1 Prepare the incident dataset for regression. Inspect missing values, impute them appropriately, encode categorical predictors, create an 80/20 training and test split and standardise the predictor variables using the training data.

(8 Marks)

3.2 Implement Linear Regression using Gradient Descent. Your solution must initialise the model parameters, compute predictions and mean-squared-error cost, calculate gradients, update the parameters iteratively and retain the cost history. Use a learning rate and number of iterations that achieve convergence. Plot the training cost over the iterations.

(14 Marks)

3.3 Evaluate the fitted model on the test set using MAE, RMSE and R^2. Interpret the three metrics in the context of restoration-time prediction.

(8 Marks)

[Subtotal: 30 Marks]

End of Question 3

Question 4 — 15 Marks

Study the scenario and complete the question(s) that follow:

Smart-Building Energy Profile Segmentation

A property-management company monitors 178 commercial buildings. The buildings differ in indoor conditions, HVAC runtime, occupancy, water use, building-envelope characteristics, peak electrical demand, ventilation and annual energy use.

The company wants to discover naturally occurring operational profiles so that energy-efficiency interventions can be tailored to different groups of buildings. No predefined class label is available.

Use K-Means clustering on smart_building_profiles.csv. Appendix C describes the fields.

4.1 Exclude Building_ID, standardise all numerical variables, and use silhouette analysis for k = 2 to k = 6 to determine an appropriate number of clusters. Report the silhouette scores and your selected k.

(5 Marks)

4.2 Fit the final K-Means model using the selected k. Report the number of buildings in each cluster. Reduce the standardised data to two principal components and create a scatter plot coloured by cluster.

(6 Marks)

4.3 Profile the clusters using the cluster-centre values in the original measurement units. Give a concise operational interpretation of the three groups.

(4 Marks)

[Subtotal: 15 Marks]

End of Question 4

Appendix

Appendix A: machine_service_risk.csv

Dataset used in Questions 1 and 2. The target variable is Major_Service_Required.

Field descriptions

Column	Description
Asset_ID	Unique pump-asset identifier. Not a predictive input.
Equipment_Age_Months	Age of the asset in months.
Service_Contract	Service-contract category: Standard or Extended.
Operating_Zone	Operational service zone: Central, North, Coastal or Remote.

Column	Description
Maintenance_Plan	Maintenance programme level: Basic, Enhanced or Premium. Contains missing values.
Average_Load_Tonnes	Average mechanical load handled by the pump.
Dust_Exposure_Level	Recorded environmental dust exposure: Low or High.
Weekly_Idle_Hours	Average weekly idle time.
Daily_Operating_Hours	Average daily hours of operation.
Lubrication_Quality_Score	Lubrication-quality indicator on an approximately 0-10 scale.
Cooling_Efficiency_Score	Cooling-system efficiency indicator.
Similar_Model_Failure_History	Whether similar units previously experienced major failure (0/1).
Hydraulic_Fault_History	Historical hydraulic fault indicator (0/1).
Electrical_Fault_History	Historical electrical fault indicator (0/1).
Control_System_Fault_History	Historical control-system fault indicator (0/1).
Impact_Damage_History	Previous impact-damage indicator (0/1).
Overpressure_Event_History	Previous overpressure-event indicator (0/1).
Peak_Hydraulic_Pressure_Bar	Recorded peak hydraulic pressure.
Idle_Hydraulic_Pressure_Bar	Recorded idle hydraulic pressure.
Monthly_Energy_Use_kWh	Monthly energy-use indicator.
Average_Bearing_Temperature_C	Average bearing-temperature measure.
Coolant_Temperature_C	Coolant-temperature measure.
Lubricant_Particle_Index	Lubricant contamination/particle index.
Sensor_Health_Score	Composite sensor-health indicator.
Mechanical_Condition_Score	Composite mechanical-condition indicator.
Error_Log_Alert	Whether a recent system error alert was recorded (0/1).
Irregular_Vibration_Alert	Whether irregular vibration was detected (0/1).
Maintenance_Compliance_Score	Score reflecting completion of planned maintenance.
Sensor_Mismatch_Flag	Sensor mismatch flag (0/1).

Column	Description
Calibration_Drift_Flag	Calibration drift flag (0/1).
Noise_Anomaly_Flag	Acoustic/noise anomaly flag (0/1).
Start_Up_Failure_Flag	Recorded start-up failure flag (0/1).
Intermittent_Shutdown	Whether intermittent shutdowns were recorded: Yes/No.
Major_Service_Required	Target class: 1 = major service required; 0 = no major service required.

Appendix B: grid_restoration.csv

Field descriptions

Column	Description
Incident_ID	Unique electricity-network incident identifier.
Affected_Line_Length_km	Approximate length of affected network line in kilometres.
Fault_Category	Recorded incident/fault category.
Grid_Load_Level	Grid-load level at the time of the incident: Low, Medium or High.
Dispatch_Window	Operational dispatch period: Early Shift, Day Shift, Late Shift or Overnight.
Crew_Type	Crew configuration assigned to the incident.
Dispatch_Delay_Minutes	Minutes between incident logging and crew dispatch.
Technician_Experience_Years	Experience of the responding technician/crew lead in years.
Restoration_Time_Minutes	Continuous target: total time in minutes until service restoration.

Appendix C: smart_building_profiles.csv

Field descriptions

Column	Description
Building_ID	Unique commercial-building identifier. Exclude from clustering.
Avg_Indoor_Temperature_C	Average indoor temperature in degrees Celsius.

Column	Description
Daily_HVAC_Runtime_Hours	Average daily HVAC runtime in hours.
Occupancy_Density_per_1000m2	Average occupancy density per 1,000 square metres.
Heating_Setpoint_C	Typical heating setpoint in degrees Celsius.
Monthly_Water_Use_m3	Average monthly water use in cubic metres.
Insulation_Performance_Score	Composite building-envelope/insulation performance score.
Solar_Gain_Index	Relative solar heat-gain indicator.
Window_to_Wall_Ratio	Approximate proportion of facade area occupied by windows.
Equipment_Diversity_Index	Relative diversity of installed building equipment.
Peak_Demand_kW	Peak recorded electrical demand in kilowatts.
Lighting_Efficiency_Index	Relative lighting-efficiency indicator.
Ventilation_Rate_ACH	Ventilation rate measured as air changes per hour.
Annual_Energy_Use_kWh_m2	Annual energy-use intensity in kWh per square metre.

End of Appendix